

Lab Experiment 3: Using Single-User Mode to Gain Access to a System

Objective

- To understand what **single-user mode** is.
- To learn how to boot into **single-user mode** in Ubuntu running on VirtualBox.
- To use single-user mode for system maintenance, such as resetting a forgotten root password or repairing configurations.

Background

In Linux, **single-user mode** is a run level that boots the system into a minimal environment where only the root user has access. It is often used for:

- System recovery.
- Resetting passwords.
- Repairing misconfigured files.
- Troubleshooting boot issues.

Since no other users or services are running, it provides full administrative control. In this lab, we will simulate entering single-user mode on Ubuntu running inside VirtualBox and perform administrative tasks.

Steps to Perform

Step 1: Power on the Virtual Machine

1. Open **Oracle VirtualBox Manager**.
2. Select your **Ubuntu VM**.
3. Click **Start** to boot.

Step 2: Access the GRUB Menu

1. When the VM starts booting, immediately press and hold **Shift** (or press **Esc** repeatedly) to access the **GRUB bootloader menu**.
2. If you don't see GRUB, reboot and try again.

Step 3: Edit Boot Parameters

1. In the GRUB menu, highlight the **Ubuntu** entry (default).
2. Press **e** to edit the boot parameters.
3. Locate the line that starts with:

```
linux /boot/vmlinuz-...
```

and contains `ro quiet splash`.

4. Replace `ro quiet splash` with:

```
rw init=/bin/bash
```

`rw` means mounts the root file system in read/write mode.

`init=/bin/bash` means starts the system directly in a bash shell as root.

Step 4: Boot into Single-User Mode

1. Press **Ctrl + X** or **F10** to boot with the modified parameters.
2. The system will boot directly into a **root shell** without a password prompt.

Step 5: Perform Administrative Tasks

Here are **20 administrative tasks** to try in single-user mode:

1. Check who you are (confirm root):

```
whoami
```

2. Check your current directory:

```
pwd
```

3. List files in root directory:

```
ls /
```

4. Create a new directory inside root directory:

```
mkdir /practice
```

5. Create an empty file:

```
touch /practice/file1.txt
```

6. Write text into a file:

```
echo "Hello Linux" > /practice/file1.txt
```

7. View contents of the file:

```
cat /practice/file1.txt
```

8. Create another file quickly:

```
nano /practice/file2.txt
```

(Type something, save with **Ctrl+O**, exit with **Ctrl+X**.)

9. Remove a file:

```
rm /practice/file1.txt
```

10. Create a new user:

```
adduser student1
```

11. Set/change password of the new user:

```
passwd student1
```

12. Change the root password:

```
passwd
```

13. List all users in the system:

```
cat /etc/passwd
```

14. Delete a user:

```
deluser student1
```

15. Check available disk space:

```
df -h
```

16. Check memory usage:

```
free -h
```

17. Check system hostname:

```
hostname
```

18. Change hostname (temporary):

```
hostname mysystem
```

19. See running processes:

```
ps -e
```

20. Reboot the system forcefully:

```
reboot -f
```

Viva Questions with Answers

Questions

Q1: What is single-user mode used for?

A: It is used for system recovery, resetting passwords, and troubleshooting.

Q2: How do you enter the GRUB menu in VirtualBox?

A: By pressing and holding **Shift** (or repeatedly pressing **Esc**) while booting.

Q3: Why do we use `rw init=/bin/bash`?

A: To mount the root filesystem as writable and start directly in a bash shell.

Q4: What command creates a new user?

A:

```
adduser username
```

Q5: How do you set a password for a user?

A:

```
passwd username
```

Q6: How do you change the current user password?

A:

```
passwd
```

Q7: What command lists all users in the system?

A:

```
cat /etc/passwd
```

Q8: How do you remove a user?

A:

```
deluser username
```

Q9: What command checks disk space?

A:

```
df -h
```

Q10: What command reboots the system?

A:

```
reboot -f
```

Detailed Questions

Q1: What is single-user mode in Linux?

A: It is a mode where the system boots into a minimal environment with root access, usually used for system recovery or maintenance.

Q2: How is single-user mode different from multi-user mode?

A: In single-user mode, only the root user is active and no services are started, while in multi-user mode multiple users can log in and services (like networking) are running.

Q3: Why do we use `rw init=/bin/bash` in the GRUB configuration?

A:

- `rw` allows the root filesystem to be writable.
- `init=/bin/bash` tells the system to start directly into a bash shell instead of the default `init` process.

Q4: How can single-user mode be useful for system administrators?

A: It allows resetting forgotten root passwords, fixing corrupted files, repairing file systems, and troubleshooting boot problems.

Q5: What is GRUB and why do we use it here?

A: GRUB (GRand Unified Bootloader) is the bootloader for Linux. We use it to modify boot parameters and enter single-user mode.

Q6: What command is used to reset the root password in single-user mode?

A:

```
passwd root
```

Q7: What command is used to reboot the system back to normal mode?

A:

```
exec /sbin/init
```

or

```
reboot -f
```